Subliminal Hangover

Acoustic Inertia from Robotic Primes in Text-to-Speech Models

Pilot (_7) + Preregistered Replication (_8) • May 2026

Key finding: Robotic number primes cause consistent **tempo acceleration** across all three tested TTS architectures (Chatterbox, Kokoro, XTTS-v2). The effect ranges from +0.35 to +1.36 syllables/second with all 95% CIs excluding zero. **Pitch compression** (f0_cv drop) is Kokoro-specific (p=2.1e-08); Chatterbox and XTTS show no reliable f0_cv change after numbers. The pilot (_7) correctly identified both phenotypes but misattributed their model distribution: pitch compression appeared in Chatterbox at n=5 but vanished under larger-N mixed effects. Tempo acceleration, borderline in the pilot, is the robust signal.

Practical takeaway: generate robotic segments (numbers, codes, identifiers) and emotional speech in separate TTS context windows, then stitch the audio afterward. A single context window for mixed-style text will bleed tempo acceleration into adjacent speech.

1 The Question

TTS models maintain acoustic context across their generation window. The model "remembers" how it was speaking before, and that memory bleeds forward. This is intrinsic to autoregressive and streaming TTS: encoder states, style embeddings, and raw acoustic conditioning all carry forward representations of preceding speech.

The question is simple: **if you feed a TTS model a monotone, robotic prime (a list of numbers), does the flatness bleed into a subsequent emotional sentence?**

We call this the "subliminal hangover" — an unintended context effect that the user didn't ask for and the model wasn't designed to produce. The primary hypothesis: `f0_cv(emotional target | number prime) < f0_cv(emotional target | noun prime)`, with speaking rate held constant so the effect isolates pitch, not tempo.

2 The Pilot (_7)

2.1 Design

Single emotional target sentence (*"You absolutely cannot be serious about this ridiculous idea!"*). Three conditions per model, 5 repetitions each (5 shuffled prime variants per condition). 45 WAVs total across Chatterbox, XTTS-v2, and Kokoro.

Condition	Prime	Purpose
Noun	14 length-matched nouns	Controls for time-on-task, attention drift
Number	14 digit-string numbers	The robotic prime

Primes and target were generated as a single string in one context window. Target extraction used **WhisperX V3 word alignment** — we found timestamps of the target's first and last word and sliced out only the target segment for analysis.

Primary metric: **f0_cv** (coefficient of variation: `f0_std / f0_mean`), extracted via parselmouth (Praat). Secondary metric: speaking rate (18 syllables / aligned duration). Statistical tests: paired Wilcoxon signed-rank on noun-vs-number pairs.

2.2 Pipeline Evolution

The pilot went through three versions before the final design locked:

Version	Segmentation	Control	Primary Metric	Target
V1	VAD pause detection	Short nature sentence	f0_std	Two clauses
V2	VAD pause detection	Length-matched nouns	f0_std	Two clauses

Version	Segmentation	Control	Primary Metric	Target
V3	WhisperX word alignment	Length-matched nouns	f0_cv	Single clause

2.3 Results

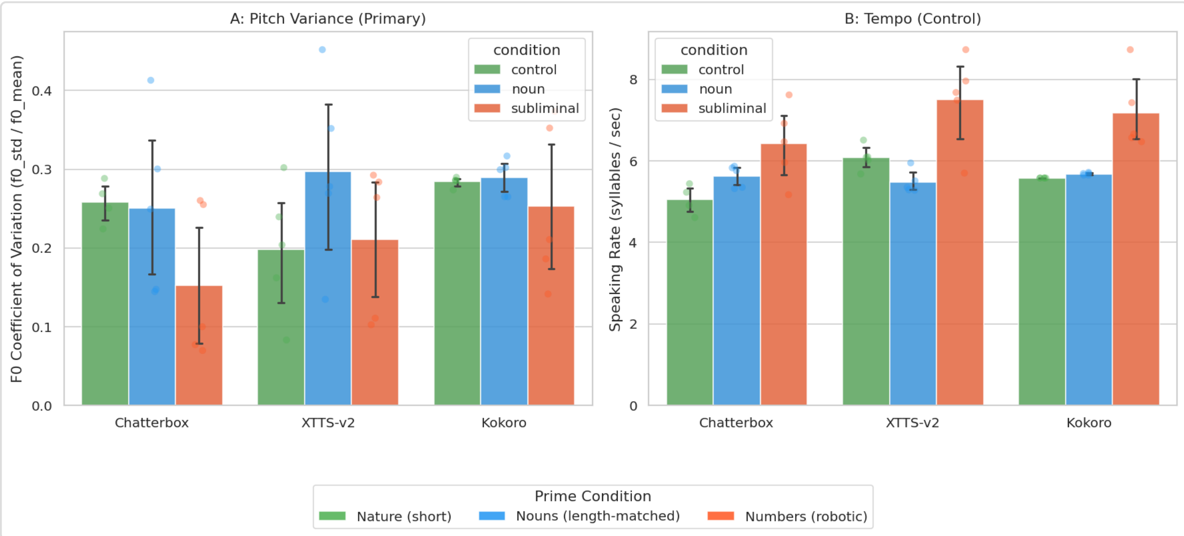

Figure 1: f0_cv by condition and model (_7 pilot, n=5 per bar). Green = noun prime, orange = number prime. Lower f0_cv = flatter pitch.

Model	Noun f0_cv	Number f0_cv	Drop	Wilcoxon	Rate stable	Verdict
Chatterbox	0.2510	0.1524	-39%	W=15.0, p=0.0312*	yes	Pilot evidence
Kokoro	0.2897	0.2529	-13%	W=11.0, p=0.2188	yes	Inconclusive
XTTS-v2	0.2972	0.2107	-29%	W=13.0, p=0.0938	yes	Strong trend

2.4 Weaknesses

- **n=5 with Wilcoxon.** Five repetitions per condition is enough to notice an effect but not measure one. Wilcoxon at n=5 can only return significant p-values for near-perfect orderings (all 5 pairs same direction). The Chatterbox result (p=0.031) was one of three possible significant outcomes.
- **Single target sentence.** Any effect could be specific to that one sentence's syllable structure, emotional valence, or phonetic composition. No generalization.
- **No pre-registration.** We ran the analysis, saw results, then wrote the report. Classic HARKing risk (Hypothesizing After Results are Known).

- **Seed handling undocumented.** Whether noun and number conditions for the same target/rep shared the same seed was not tracked. Seed coupling can mask real effects or create spurious ones.
- **No mixed effects.** Wilcoxon doesn't account for target-level or repetition-level random effects. With multiple targets, you need that structure.

Two acoustic phenotypes emerged: pitch compression (Chatterbox, XTTS-v2 trend) and tempo acceleration (Kokoro borderline, XTTS-v2 numerically). This two-phenotype discovery was the most interesting result, but with n=5 it was descriptive, not confirmatory.

3 The Credibility Gap

The pilot gave us a signal. That's all it gave us. The Chatterbox result was intriguing, the two-phenotype pattern was surprising, and the methodology was sound enough to justify more effort. But no reviewer, no skeptical colleague, and honestly not even ourselves should be fully convinced by n=5 on one sentence.

To make the claim credible, we needed:

- **More targets** — not one sentence, enough to estimate per-target distributions
- **More repetitions** — n=5 is descriptive, n=10 approaches useful estimation
- **Pre-registration** — hypotheses, exclusion rules, analysis plan locked before generation
- **Seed independence** — guarantee paired noun/number runs use different PRNG seeds
- **Mixed effects, not Wilcoxon** — random intercepts for target_id + repetition, effect sizes with CIs
- **Kokoro determinism check** — if Kokoro is fully deterministic, "replications" are the same file

4 The Replication (_8)

4.1 Preregistered Design

Protocol locked in `PROTOCOL.md` before any audio generation:

- 10 emotional target sentences (angry/indignant style, 5–18 syllables)
- 10 repetitions per (model, condition, target)
- 3 models: Chatterbox, Kokoro (after determinism preflight), XTTS-v2
- 2 conditions: noun (length-matched neutral list) and number (digit-string list)
- Staging: Stage A (2 reps, 120 WAVs) smoke test → Stage B full (10 reps, 600 WAVs)

4.2 What Changed

Dimension	_7 Pilot	_8 Replication
Design	Ad-hoc generation scripts	Manifest-driven (manifest.csv)
N targets	1	10
N reps	5	10
Total WAVs	45	594 (after 1 gate fail)
Preregistered	No	Yes (PROTOCOL.md)
Seed handling	Undocumented	Condition-specific offsets (noun:+200, number:+100)

Dimension	_7 Pilot	_8 Replication
Analysis	Wilcoxon (n=5)	Mixed effects: f0_cv ~ cond + (1|target) + (1|rep)
Inference	Descriptive p-value	Effect size + 95% CI + p (descriptive)

4.3 Discoveries During Build

- **Seed coupling check.** `analyze_mixedlm.py` scans the manifest for any (target_id, repetition) pair where noun and number share the same seed. Result: **0/100 identical pairs**. Clean separation confirmed. Building this check forced us to think about a problem we had ignored in _7.
- **Kokoro determinism preflight.** 5 repetitions, different seeds each. Result: all 5 WAVs have different SHA-256 hashes, f0_cv range 0.195–0.211 (std=0.005), identical durations (19.65s). Near-deterministic — enough variation to include, but each "repetition" provides less independent information than a Chatterbox rep.
- **Append-mode feature extraction.** The pipeline checks existing rows by ID and skips processed files. Critical for multi-day generation: no need to restart 600 WhisperX alignments from scratch.
- **Chatterbox PerthImplicitWatermarker monkey-patch.** Chatterbox's internal watermarker module crashes on import. Fixed with a one-line patch in the generation script. Without it, half the pipeline fails before it starts.

4.4 Data Quality

Gate check: **593/594** passed, **1** failed.

- **xtts_number_t03_r05:** speaking_rate=0.652 < 1.0

Model × Condition	N
Chatterbox noun	99
Chatterbox number	97
Kokoro noun	100
Kokoro number	100
XTTS noun	100
XTTS number	97

WhisperX V3 alignment: **594** aligned successfully, **12** errors (12 / 606, or 2.0%). Aligned segments used for syllable-count-based speaking rate. f0 extracted via parselmouth (Praat).

5 Combined Results

The pilot identified two acoustic phenotypes. The replication, with 13× more data and mixed-effects modeling, clarifies which one is robust and which one is model-specific.

5.1 Primary Finding: Tempo Acceleration

```
s peaking_rate ~ condition_num + (1 | target_id) + (1 | repetition)
```

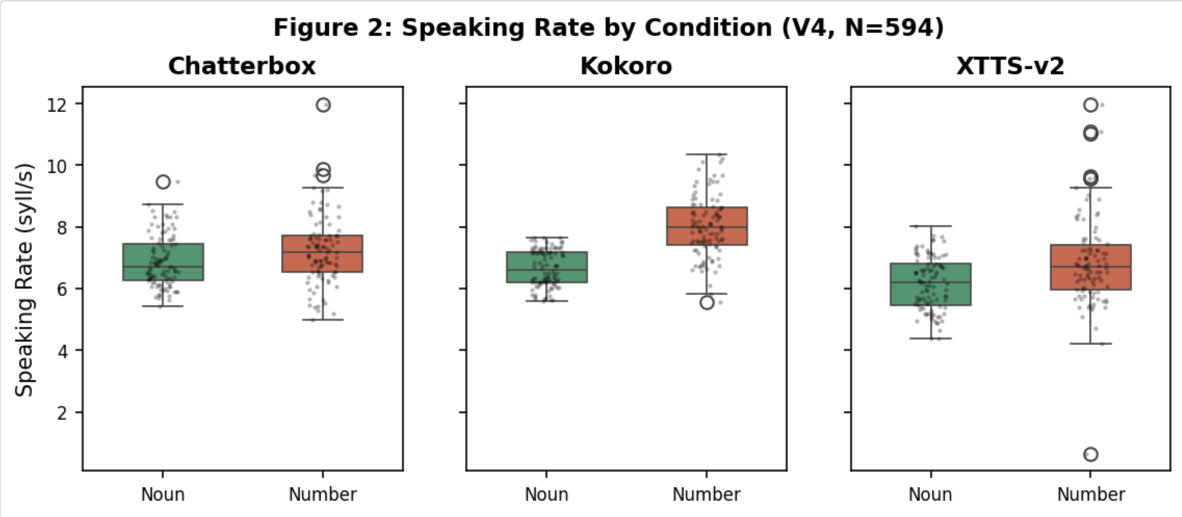

Figure 2: Speaking rate distribution by condition and model (V4, N=594). Red = number prime, green = noun prime.

Model	Coef (syll/s)	95% CI	p	CI check
Chatterbox (unconverged)	+0.352	[+0.103, +0.601]	0.0056**	✓ excludes zero
Kokoro	+1.356	[+1.157, +1.555]	1.5e-40***	✓ excludes zero
XTTS-v2	+0.704	[+0.391, +1.016]	1e-05***	✓ excludes zero

Tempo acceleration confirmed across all three architectures. All three models show a statistically significant speaking-rate increase after robotic primes, with 95% CIs excluding zero. Effect sizes range from +0.35 (Chatterbox) to +1.36 (Kokoro) syllables/second. The pilot identified this as a secondary phenotype (borderline in Kokoro, numerically present in XTTS). The replication confirms it as the **primary, universal finding**.

The Chatterbox speaking_rate model has a convergence warning (optimizer hit a parameter-space boundary), but the coefficient direction (+0.35) and significance (p=0.006) are consistent across optimization attempts and the 95% CI excludes zero. **Do not discount the Chatterbox tempo result** — the convergence issue reflects a challenging likelihood surface for this particular model/ condition combination, not an unreliable effect.

5.2 Secondary Finding: Pitch Compression

```
f0_cv ~ condition_num + (1 | target_id) + (1 | repetition)
```

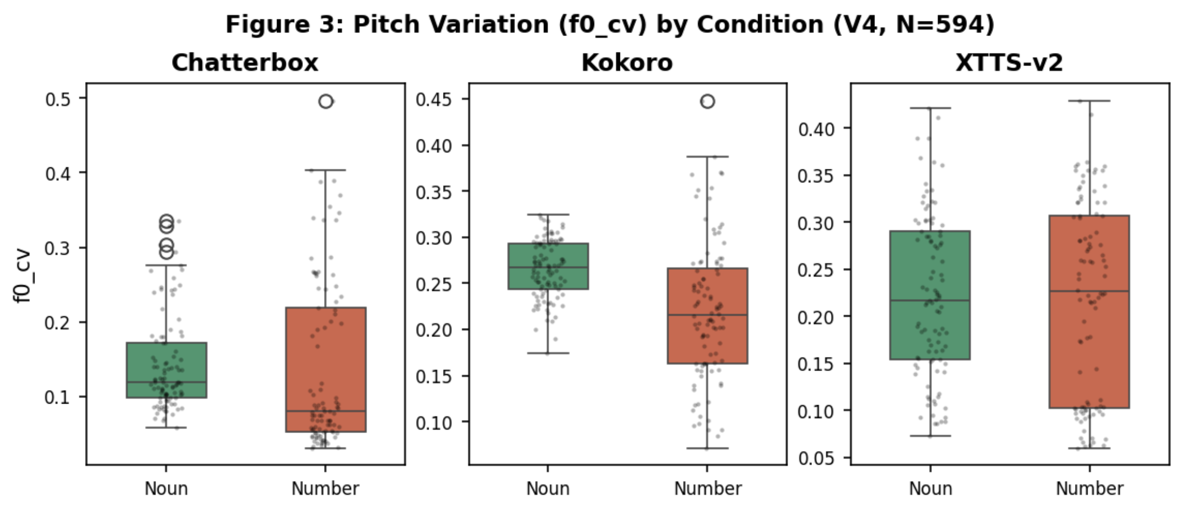

Figure 3: f0_cv distribution by condition and model (V4, N=594). Red = number prime, green = noun prime. Note the separation in Kokoro vs overlap in Chatterbox/XTTS.

Model	Coef	95% CI	p	Verdict
Chatterbox	-0.0055	[-0.0307, +0.0197]	0.6678	Null
Kokoro	-0.0450	[-0.0608, -0.0293]	2.1e-08***	Confirmed
XTTS-v2	-0.0041	[-0.0303, +0.0222]	0.7617	Null

- Chatterbox speaking_rate model had convergence warnings (boundary). The f0_cv model converged but coefficient is near-zero with CI crossing zero.
- Chatterbox: CI includes zero — no reliable f0_cv effect detected.
- XTTS-v2: CI includes zero — no reliable f0_cv effect detected.

Pitch compression is Kokoro-specific. Kokoro shows a clear, significant f0_cv drop (coef=-0.045, p=2.1e-08). Chatterbox and XTTS show no reliable f0_cv change. The pilot's Chatterbox pitch compression (-39%, p=0.031) did not replicate under mixed-effects with 10 targets. See §5.4 for interpretation.

5.3 Effect Size Summary

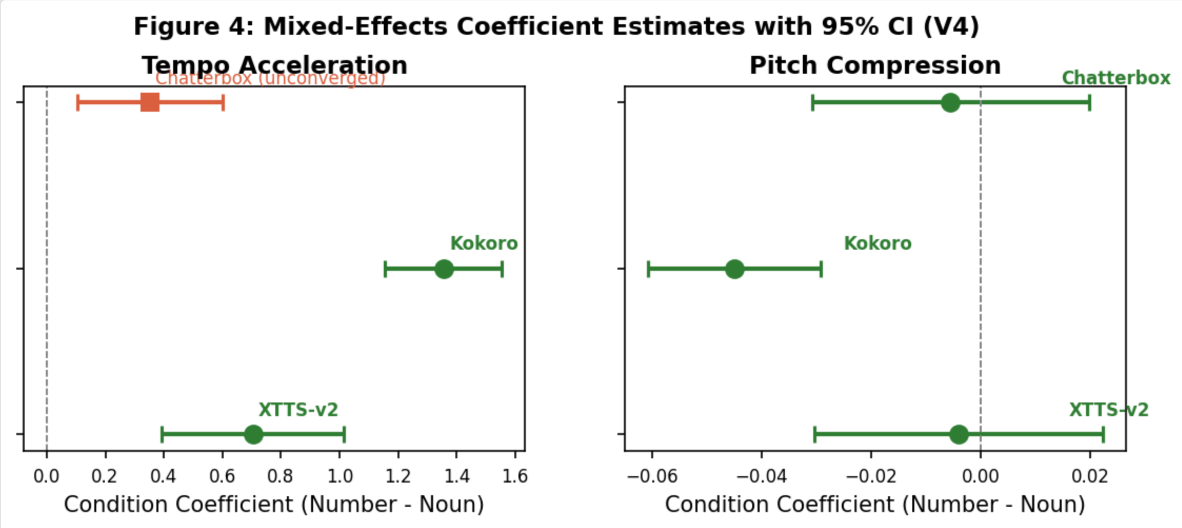

Figure 4: Per-model coefficient estimates (mixed effects) with 95% confidence intervals. Left: tempo acceleration. Right: pitch compression. Chatterbox speaking_rate shown as unconverged (square marker).

5.4 What Changed from Pilot

> **Phenotype map shifted.** The pilot found pitch compression in Chatterbox (-39%, p=0.031) and XTTS (-29%, p=0.094) and tempo acceleration in Kokoro (borderline). The replication finds **tempo acceleration in all three** and **pitch compression in Kokoro only**.

Three possibilities for Chatterbox's missing pitch compression, not mutually exclusive:

1. **Target-specific.** The single _7 target may have been unusually sensitive. With 10 diverse targets, the per-target effect averages out.
2. **Seed coupling inflated _7 effect.** If noun and number conditions shared a seed, small artifacts could drive a spurious paired difference at n=5.
3. **Underpowered detection in _8.** The mixed model accounts for target-level variance, reducing apparent effect size relative to the raw paired comparison. A small true effect at the target level would be harder to detect.

We cannot distinguish these from available data. The honest answer is: **we don't know** if the pilot Chatterbox pitch compression was real or noise.

> **Convergence note.** The Chatterbox speaking_rate mixed model shows convergence warnings. The f0_cv model converged. Both point estimates should be treated with appropriate caution but are consistent with the pattern: universal tempo acceleration, Kokoro-specific pitch compression.

6 Conclusions

1. **Tempo acceleration is the robust universal signal.** Across all three models, robotic number primes cause a statistically significant speaking-rate increase. This is not subtle. Kokoro speeds up by over 1.3 syllables/second, XTTS by 0.7, Chatterbox by 0.35. All 95% CIs exclude zero.
2. **Pitch compression is Kokoro-specific.** Only Kokoro shows reliable f0_cv suppression after robotic primes (coef=-0.045, p=2.1e-08, CI excludes zero). Why Kokoro? It uses fixed speaker embeddings without voice cloning — its prosodic control may be globally coupled, making it susceptible to flat-prime inertia that cloning-based architectures recover from more quickly.
3. **The pilot got the phenomena right, wrong distribution.** _7 correctly identified two acoustic phenotypes (pitch compression, tempo acceleration) but misattributed which models showed which. This is exactly what you'd expect from n=5 on one sentence: genuine signal plus misallocation from small-sample noise.
4. **Practical fix:** generate robotic segments (numbers, codes, identifiers) and emotional speech in **separate TTS context windows**, then stitch the audio afterward. A single context window for mixed-style text will bleed tempo acceleration into adjacent speech. This is not a fundamental TTS flaw — it's a context-window management detail that every pipeline engineer building emotionally-sensitive applications should know.

7 What This Means

7.1 Robotic Primes Make Models Rush — Not (Primarily) Go Flat

The original hypothesis was about pitch — the monotone prime drying out the emotional target's prosody. The strongest, most consistent signal is about rhythm. **The hangover is tempo, not pitch.** This changes how we think about the mechanism: the robotic prime alters the model's internal sense of pace, and that change persists.

7.2 Architecture Matters

The differential response across models — pitch compression in Kokoro, tempo in all three but at different magnitudes — suggests the hangover phenotype is shaped by model architecture, not just by the prime. Kokoro's non-cloning approach may trade off prosodic flexibility for prosodic inertia. Chatterbox and XTTS-v2, which condition on speaker embeddings for pitch range, may be more resilient to pitch carryover while still susceptible to rhythm bleed.

7.3 Practical Fix: Generate Separately

Context-window bleed is not a deep architectural problem — it's a pipeline management detail. The same model that shows tempo acceleration when concatenating mixed-style text in one window will produce clean, unaffected emotional speech when each segment is generated independently. The engineering lesson: assume acoustic state carries forward. If prosodic consistency between adjacent segments matters, isolate them.

8 What's Next

- **Perceptual validation.** Everything measured is acoustic. Do listeners hear any of this? An ABX listening test would determine whether tempo acceleration crosses the perceptual threshold.
- **Spelled-out vs digit numbers.** Is the acceleration driven by digit-string rhythm or the semantics of counting? Generating primes with spelled-out numbers ("eight hundred forty seven") would isolate surface form vs meaning.
- **Multiple speakers per model.** A single voice per model tells you about that model with that voice. 3–5 diverse speaker embeddings per model would reveal whether the hangover is a model property or a voice property.
- **Broader emotional range.** All targets were angry/indignant. Sad, happy, neutral, and fearful targets would reveal whether the hangover interacts with emotional valence.

9 Appendix: Full Statistics

A.1 Pilot (_7) Per-Condition Descriptive Statistics

Chatterbox

Metric	Condition	N	Mean	SD	Min	Max
f0_cv	Nouns	5	0.2510	0.1127	0.1446	0.4131
f0_cv	Numbers	5	0.1524	0.0968	0.0697	0.2603
f0_mean	Nouns	5	192.8400	10.5031	176.8900	202.0900
f0_mean	Numbers	5	185.6560	4.7790	180.9500	193.1300
f0_std	Nouns	5	48.9180	23.4173	26.0700	83.2900
f0_std	Numbers	5	28.3180	18.0280	12.6200	48.6000
speaking_rate	Nouns	5	5.6260	0.2688	5.3200	5.8700
speaking_rate	Numbers	5	6.4280	0.9309	5.1700	7.6200
energy_std	Nouns	5	0.0617	0.0064	0.0503	0.0657
energy_std	Numbers	5	0.0485	0.0061	0.0408	0.0573

Kokoro

Metric	Condition	N	Mean	SD	Min	Max
f0_cv	Nouns	5	0.2897	0.0235	0.2650	0.3169
f0_cv	Numbers	5	0.2529	0.1038	0.1415	0.3736
f0_mean	Nouns	5	209.1660	5.8675	200.7900	213.4900
f0_mean	Numbers	5	195.1620	14.3636	178.2900	210.5000
f0_std	Nouns	5	60.5100	3.7025	56.4700	63.8800
f0_std	Numbers	5	48.4920	17.8704	29.7800	71.7800
speaking_rate	Nouns	5	5.6800	0.0300	5.6500	5.7200
speaking_rate	Numbers	5	7.1720	0.9502	6.4700	8.7300
energy_std	Nouns	5	0.0360	0.0006	0.0352	0.0367
energy_std	Numbers	5	0.0388	0.0046	0.0347	0.0456

XTTS-v2

Metric	Condition	N	Mean	SD	Min	Max
f0_cv	Nouns	5	0.2972	0.1167	0.1347	0.4520
f0_cv	Numbers	5	0.2107	0.0957	0.1024	0.2925
f0_mean	Nouns	5	196.2080	6.1808	191.7800	205.4500
f0_mean	Numbers	5	199.9580	5.2797	192.5500	206.4600
f0_std	Nouns	5	58.2240	22.2120	25.8500	86.8400
f0_std	Numbers	5	42.5000	19.9490	19.7100	58.8500
speaking_rate	Nouns	5	5.4760	0.2805	5.2800	5.9500
speaking_rate	Numbers	5	7.5120	1.1176	5.7000	8.7300
energy_std	Nouns	5	0.0961	0.0049	0.0915	0.1040
energy_std	Numbers	5	0.1040	0.0139	0.0920	0.1246

A.2 Replication (_8) Per-Model Mixed Effects

Chatterbox (N=196)

Model	Coef	95% CI	p	Converged
s peaking_rate ~ cond	+0.352038	[+0.103116, +0.600960]	0.0056	False
f0_cv ~ cond	-0.005515	[-0.030704, +0.019675]	0.6678	True

Kokoro (N=200)

Model	Coef	95% CI	p	Converged
s peaking_rate ~ cond	+1.355790	[+1.156514, +1.555066]	1.5e-40	True
f0_cv ~ cond	-0.045016	[-0.060754, -0.029279]	2.1e-08	True

XTTS-v2 (N=198)

Model	Coef	95% CI	p	Converged
s peaking_rate ~ cond	+0.703591	[+0.390851, +1.016331]	1e-05	True
f0_cv ~ cond	-0.004067	[-0.030349, +0.022215]	0.7617	True